

Canonical initialization: continuation certificate and reachability audit

The initialized canonical reversal theorem remains open. This supplement verifies a twelvefold larger sufficient learned-state entry region, audits the exact inputs and the local proof chain, and records the initialized continuation and first-event checks. It neither asserts a high-probability result nor certifies the empirical timing lag.

1 Verified canonical continuation

1.1 Model, inputs, and arithmetic scope

Write $\theta = (U, W)$, with all input rows followed by all output rows, $f_\theta(x) = \sum_i w_i(u_i^T x)_+$ and $\mathcal{L} = \frac{1}{2}\mathbb{E}_n\|x - f_\theta(x)\|^2$. The full training residual is $e = x - f_\theta(x)$, and $F = J^*e$ denotes a compatible gradient field. The adjoint uses the empirical inner product. The bounds below cover all compatible selectors; no global uniqueness is assumed. Global Clarke flows and the compatible representation are those of the supplied revised appendix’s nonsmooth-flow lemma.

Put $r = \sin(\pi/36)$, $k = \cos(\pi/36)$, $v = (r, -1)$, and $M = \sum_{i \in I} w_i u_i^T$. For $Y = e_1^T M v - r$, the empirical quantity is

$$Q_G = Y(r\dot{m}_{11} - \dot{m}_{12}) + r\dot{m}_{21} - \dot{m}_{22} = \nabla \Phi_I(\theta)^T F(\theta), \quad \Phi_I = \frac{1}{2}Y^2 + e_2^T M v.$$

Its output gradient is $\ell = (Y, 1)$, so $g_{u_i} = \mathbf{1}_I v(w_i^T \ell)$ and $g_{w_i} = \mathbf{1}_I \ell(u_i^T v)$. The unit central probe is $(r, -k)$; no k factors are inserted into Q_G . The canonical script uses exact field evaluation. This is distinct from the finite-difference estimator used by the nine-setting study.

The proof inputs and code are hashed in the accompanying manifest. The repository commit is

`ae16f9c0e4e51269c624ffb196184568f2cfe6d9`.

The exact archived initial parameter rows are preserved; freshly evaluating the trigonometric initialization changes some entries and is not used. The supplied binary dataset exactly matches the committed seeded generator under the recorded runtime. The historical state archive contains neither raw X nor an independent X hash, so historical bitwise dataset identity is not independently established. The theorem is about the supplied binary instance, not an iid Gaussian event.

Arithmetic bounds use midpoint-radius intervals, outward IEEE rounding bounds for sums and dot products, and interval transcendental evaluation. They assume standard round-to-nearest operations without unsafe fast-math. All numerical tube radii are verified by strict inequalities; no integrator accuracy claim is accepted without a reference-defect bound.

1.2 A larger certified entry region

Let X be the exact supplied binary dataset with $N = 4000$, $h = 200$, and the canonical generation parameters $(\mu_1, \mu_2, \sigma) = (3, 2, 0.15)$. Let $\hat{\theta}_{3.4}$ be the supplied reference state. The fixed 71-neuron cohort I is selected by the archived in-distribution rule at $t = 4$, not by OOD outcomes. Put

$$\xi_\alpha = (\cos \alpha, \sin \alpha), \quad \mathcal{I} = \{\alpha : |\alpha + 17\pi/36| < 10^{-6}\}.$$

The theorem refers to global Clarke gradient flows of the empirical loss; it applies to every such flow satisfying its entry condition. No global uniqueness assumption is needed. Compatible selector fields are used as an overapproximation of the Clarke inclusion, including at zero preactivation.

Theorem 1.1 (Canonical continuation from a learned-state neighborhood). *If*

$$\|\theta(3.4) - \hat{\theta}_{3.4}\|_2 \leq 1.2 \cdot 10^{-5}, \quad (1)$$

then, uniformly for $\alpha \in \mathcal{I}$,

$$\begin{aligned} E(\xi_\alpha, 3.4) - E(\xi_\alpha, 3.7) &> 1.2 \cdot 10^{-4}, \\ E(\xi_\alpha, 3.9) - E(\xi_\alpha, 3.7) &> 4 \cdot 10^{-5}. \end{aligned}$$

The error has an interior minimum on $[3.4, 3.9]$. At entry the cohort has $m_{11} > 0.9323$, input angles below 15° , and output angles below 20° . For the central probe, both Q_G and the full OOD rate are negative at entry and positive at 3.9 for all compatible training selectors. The strong-cluster contribution to the full OOD rate is negative at both endpoints, the weak-cluster contribution is positive, and the latter dominates at the late endpoint. Moreover, almost everywhere, for all probes in the sector,

$$\dot{E} < -0.0012 \quad \text{on } [3.4, 3.40001], \quad \dot{E} > 0.00040 \quad \text{on } [3.9, 3.90001].$$

The sector has width $2 \cdot 10^{-6}$ radians. Neither an initialized entry guarantee nor a unique continuous zero of Q_G is asserted.

Proof. The certificate applies the one-sided tube inequality and the verified-step criterion below to 2500 reference intervals of exact length $1/5000$. Every reference coefficient is an exact stored binary number; its numerical integration provenance is not a proof premise. The uniform interval bounds are independent of the chosen initial error within the proposed tubes. An independent scalar replay with upward-rounded initial radius $1.2 \cdot 10^{-5}$ verifies every strict tube inequality. At the snapshots,

$$\|\theta(3.7) - \hat{\theta}_{3.7}\| < 2.828 \cdot 10^{-5}, \quad \|\theta(3.9) - \hat{\theta}_{3.9}\| < 4.251 \cdot 10^{-5}.$$

The full step ledger is `WIDE_LOCAL_FLOW_ENCLOSURE.csv`. Static bounds use radii $2 \cdot 10^{-5}$, $3.5 \cdot 10^{-5}$ and $4.6 \cdot 10^{-5}$ at the three snapshots. On their respective balls,

$$\begin{aligned} E(\xi_0, 3.4) &\in [0.48314641, 0.48315250], \\ E(\xi_0, 3.7) &\in [0.48298853, 0.48299930], \\ E(\xi_0, 3.9) &\in [0.48305831, 0.48307257], \quad \xi_0 = \xi_{-17\pi/36}. \end{aligned}$$

These displayed intervals are enlarged from the computed enclosures. The full-precision central gaps exceed 0.00014712 and 0.00005902. On the unit circle the error is Lipschitz in the probe with constant $(1 + \|\theta\|^2/2)^2 < 9.084$. Transporting two errors through an angular change below 10^{-6} costs less than $1.8168 \cdot 10^{-5}$. The resulting verified sector gaps are greater than 0.00012895 and 0.00004085, proving the stated margins and the interior minimum.

The endpoint interval bounds for 10^3 times the exact rates are:

Statistic	contribution	$t = 3.4$	$t = 3.9$
Q_G	total	$[-1.140, -0.960]$	$[0.630, 1.029]$
$\dot{E}$	total	$[-1.379, -1.215]$	$[0.406, 0.771]$
$\dot{E}$	strong cluster	$[-2.468, -2.320]$	$[-1.273, -0.939]$
$\dot{E}$	weak cluster	$[1.055, 1.138]$	$[1.601, 1.788]$

Each cluster uses the same full-network residual and normalization $1/N$; this is a signed field decomposition, not a retraining ablation. The same interval calculation verifies the mass and cone statements.

The speed bounds on the first and last static balls are 0.385 and 0.309. The verified endpoint radii leave enough room for a forward extension of 10^{-5} at each endpoint. A first-exit argument keeps the flow in those balls. The probe gates remain constant across each static ball and sector. Indeed their central score margins exceed $3.7575 \cdot 10^{-5}$ and $5.6601 \cdot 10^{-5}$, respectively; subtracting the parameter radius and an additional $2.01 \cdot 10^{-6}$ for sector transport still leaves positive margins. With $P < 2.01$, the OOD rate varies with the probe by at most $2(1 + P^2/2)P\|F\|\|\xi - \xi_0\| < 5 \cdot 10^{-6}$. Subtracting this cost from the full-precision rate margins gives the two uniform rate intervals. The exact static reports and arithmetic contract are supplied with the certificate. $\square$

Scope. Condition (1) specifies a parameter neighborhood of an ID-trained state; it does not assume an OOD sign. Its radius is twelve times the earlier sufficient radius. Membership of the canonical initialized flow in this neighborhood remains unproved. The result therefore does not strengthen the paper’s existing random-initialization theorem or establish that specialization alone causes reversal. The offline cohort and witness times are retrospective, and the sector is deliberately conservative.

1.3 Static sign certificates with all training-gate changes

This section concerns parameter neighborhoods, not their reachability from initialization. Its dataset consists of the exact binary numbers in the accompanying `canonical_witness_states.npz`. Rounding a Gaussian sample is not an iid Gaussian theorem.

Let y be a parameter center, $R > 0$, $P = \|y\|$, $\bar{P} = P + R$, and $D = PR + R^2/2$. A cluster index set C uses the full-dataset normalization $\|z\|_C^2 = N^{-1} \sum_{n \in C} \|z_n\|^2$. Put $\lambda_C \geq \|N^{-1} \sum_{n \in C} x_n x_n^T\|$. Write F_C for that cluster’s contribution to the full gradient field, using the *full-network residual*; $F = F_1 + F_2$. Let F_C^0 denote its polynomial extension with the training gates fixed at y . No fixed-gate hypothesis is imposed on the actual field in the ball.

For either potential used here, set $g = \nabla \Phi$, $G_0 = \|g(y)\|$. For the empirical statistic use the potential Φ_I defined above. For the full derivative, fix the actual probe-active set at y and use $\Phi(\theta) = \frac{1}{2} \|M_{\text{act}}(\theta)\xi - \xi\|^2$. This second choice is valid throughout the ball if $\min_i |u_i(y)^T \xi| > R\|\xi\|$. In either case let v be its input vector, I its fixed index set, $P_I = \|(u_i, w_i)_{i \in I}(y)\|$, and ℓ_y its output gradient. Define

$$\begin{aligned} L_C &= \lambda_C \bar{P}^2 + \sqrt{\lambda_C} (\|e(y)\|_C + \sqrt{\lambda_C} D), \\ H_g &= \|v\|^2 (P_I + R)^2 + \|v\| (\|\ell_y\| + \|v\| (P_I R + R^2/2)). \end{aligned}$$

Here are explicit bounds for the training-gate correction. Put $q_{ni} = u_i(y)^T x_n$, $s_n = \|x_n\|$, $e_n = e(y; x_n)$, and

$$\begin{aligned} d_{ni} &= (Rs_n - |q_{ni}|)_+, & p_{ni} &= \mathbf{1}_{|q_{ni}| \leq Rs_n}, \\ b_{ni} &= |w_i(y)^T e_n| + R\|e_n\| + (\|w_i(y)\| + R)Ds_n, \\ j_i^u &= N^{-1} \sum_{n \in C} p_{ni} b_{ni} s_n, & j_i^w &= N^{-1} \sum_{n \in C} d_{ni} (\|e_n\| + Ds_n), \\ z_n &= \sum_i (\|w_i(y)\| + R) d_{ni}, & B_C &= \left[\sum_i ((j_i^u)^2 + (j_i^w)^2) \right]^{1/2} + \sqrt{\lambda_C} \bar{P} \|z\|_C. \end{aligned}$$

Every neuron is retained in these quantities.

Proposition 1.2 (A static neighborhood enclosure). *For any compatible training selectors and $\|\theta - y\| \leq R$,*

$$|g(\theta)^T F_C(\theta) - g(y)^T F_C(y)| \leq R[H_g(\|F_C(y)\| + L_C R) + G_0 L_C] + (G_0 + H_g R) B_C. \quad (2)$$

This applies to the total derivative and separately to each cluster’s signed contribution. Cluster contributions always use the same full residual.

Proof. The Jacobian norm on cluster C is at most $\sqrt{\lambda_C} \|\theta\|$: apply Cauchy–Schwarz across neuron and input/output parameter coordinates before taking the empirical average. Integration along a parameter segment gives $\|f_\theta - f_y\|_C \leq \sqrt{\lambda_C} D$ and the pointwise bound $\|f_\theta(x_n) - f_y(x_n)\| \leq Ds_n$. The same bounds hold for the fixed-mask polynomial extension. Its field derivative is $-J_C^* J_C + H_{e,C}$, where the neuronwise residual Hessian has norm at most $\sqrt{\lambda_C} \|e\|_C$. Thus $\|DF_C^0\| \leq L_C$. Differentiating either potential gives $\|Dg\| \leq H_g$ on the ball. Adding and subtracting $g(y)^T F_C^0(\theta)$ proves the first term of (2).

A changed training activation differs from its fixed-mask extension by at most d_{ni} , and its selector can differ only when $p_{ni} = 1$. The pointwise residual and weight bounds give $|w_i(\theta)^T e_\theta(x_n)| \leq b_{ni}$. The direct input- and output-gradient changes are therefore bounded by j_i^u, j_i^w . The change of residual between the actual network and the fixed-mask extension has norm at most z_n . Applying the fixed-mask adjoint to this residual costs at most $\sqrt{\lambda_C} \bar{P} \|z\|_C$. Consequently $\|F_C - F_C^0\| \leq B_C$. Multiply by $\|g(\theta)\| \leq G_0 + H_g R$ to finish. Endpoint partial selectors obey the same bounds. $\square$

1.4 An enclosure through training-gate changes

This section states the real-arithmetic stability inequality used by the interval validator. A successful floating-point run is not by itself a certificate; its input and rounding enclosures must also be verified.

Write $F(\theta) = J_\theta^*(x - f_\theta)$ for the selected vector field. Fix a reference state y and a radius R . Assume that along every straight parameter segment within this ball, the smooth pieces obey

$$\max_i \|\mathbb{E}_n[e_\theta(x)x^T\alpha_i(x)]\| \leq a.$$

For each neuron and sample choose an upper bound $h_{in} \geq N^{-1}\|x_n\|(w_i^T e_\theta(x_n))_+$ throughout the ball. Set $d_{in} = |u_i(y)^T x_n|/\|x_n\|$ for nonzero samples, and

$$B_i(s) = \sum_{n:d_{in} \leq s} h_{in}, \quad 0 \leq s \leq R.$$

If the center ranges along a reference subinterval, lower bounds for the margins and uniform upper bounds for the weights can replace these quantities.

Lemma 1.3 (One-sided stability with adverse gate jumps). *Suppose $B_i(s) \leq a_i + \beta s$ for $0 \leq s \leq R$, where $a_i, \beta \geq 0$. Then for $\|z\| \leq R$,*

$$\langle z, F(y+z) - F(y) \rangle \leq (a + \beta)\|z\|^2 + \|(a_i)_i\|_2\|z\|.$$

Compatible endpoint selectors are allowed. Along an absolutely continuous reference path $y(t)$ with defect $\|F(y(t)) - \dot{y}(t)\| \leq \eta$, the error $r(t) = \|\theta(t) - y(t)\|$ therefore satisfies, while $r \leq R$,

$$D^+r \leq (a + \beta)r + \|(a_i)_i\|_2 + \eta$$

in the comparison sense (or almost everywhere away from $r = 0$).

Proof. On a smooth activation region, $DF = -J^*J + H_e$. The residual Hessian H_e is block diagonal across neurons, with each block having off-diagonal entries A_i, A_i^T , where $A_i = \mathbb{E}_n[ex^T\alpha_i]$. Thus $\lambda_{\max}(DF) \leq a$. Integrate this inequality along the segment $y + sz$, separating its finitely many training hyperplane crossings. When one gate crosses, the vector-field jump has only an input-coordinate component $\Delta F_{u_i} = N^{-1}\Delta\alpha_i(w_i^T e_x)x$; the output-coordinate component is continuous because the preactivation is zero there. Its pairing with z is at most $N^{-1}(w_i^T e_x)_+ |z_{u_i}^T x| \leq h_{in}\|z_{u_i}\|$. Only samples with $d_{in} \leq \|z_{u_i}\|$ can cross. The sum of jumps is therefore bounded by

$$\sum_i \|z_{u_i}\| B_i(\|z_{u_i}\|) \leq \|(a_i)_i\|_2\|z\| + \beta\|z\|^2.$$

Simultaneous crossings add; endpoint partial selectors satisfy the same upper bound by taking partial jumps. Segments that lie in a hyperplane contribute no conormal pairing for that gate. This proves the first claim. Take the inner product of the error equation with $\theta - y$ and divide by r when it is positive. Standard norm regularization gives the comparison at zero. $\square$

For a finite set of distances, permissible intercepts are explicit:

$$a_i(\beta) = \max \left\{ 0, \max_{0 \leq s \leq R} (B_i(s) - \beta s) \right\}.$$

The inner maximum only needs the sorted sample distances, including zero. There is no density assumption or asymptotic substitution for these counts.

Corollary 1.4 (A verified step criterion). *On a reference interval of length h , let the preceding bounds be uniform on its radius- R tube. Put $k = a + \beta$ and $\omega = \|(a_i)_i\|_2 + \eta$. If the initial error is at most $r_0 < R$ and*

$$e^{kh}r_0 + \omega \frac{e^{kh} - 1}{k} < R,$$

then the flow remains in the tube on the whole interval and its terminal error is bounded by the left side. At $k = 0$ the quotient is interpreted as h . Successive verified steps give an enclosure from the initial state.

Proof. Apply the scalar integrating-factor inequality until the first exit. Its upper bound is nondecreasing in time and strictly smaller than R at the terminal time, excluding a first exit. Iterate using the terminal upper bound as the next initial radius. $\square$

The absolute reference defect must also include any gates crossed by the reference path. The favorable sign used in Lemma 1.3 applies to *stability*, not automatically to the defect of a numerical interpolant. Using only a smooth-region defect bound would omit this term and would not constitute a certificate.

1.4.1 Cubic-reference defect used in the local validator

Let $y(s)$ be the cubic Hermite interpolant of two stored parameter states and two stored velocities, over a step of exact length $1/5000$. The stored velocities are merely coefficients of the reference polynomial; they need not be exact field values. Put $a = 1/10000$ and center the step at $s = 0$. Fix the actual training masks at $y(0)$ and denote that polynomial field by F^0 . The validator encloses

$$\begin{aligned} d_0 &= \|F^0(y) - y'\|, \\ d_1 &= \|DF^0(y)y' - y''\|, \\ d_2 &= \|D^2F^0(y)[y', y'] + DF^0(y)y'' - y'''\| \end{aligned}$$

at the midpoint. Suppose P_*, V_*, A_*, T_* bound respectively $\|y\|, \|y'\|, \|y''\|, \|y'''\|$ throughout the step, and L_* bounds $\|DF^0\|$. For the covariance bound λ ,

$$\|D^2F^0\| \leq 3\lambda P_*, \quad \|D^3F^0\| \leq 3\lambda.$$

Indeed the fixed-mask output is quadratic with second derivative norm at most $\sqrt{\lambda}$, while $\|J\| \leq \sqrt{\lambda}P_*$; differentiate $F^0 = J^*e$ twice and three times. Taylor's integral remainder therefore gives

$$\eta_{\text{sm}} = d_0 + ad_1 + \frac{1}{2}a^2d_2 + \frac{1}{6}a^3(3\lambda V_*^3 + 9\lambda P_*V_*A_* + L_*T_*)$$

as an upper bound for the fixed-mask defect on the whole step. The third derivative of the defect uses $y''' = 0$.

The true defect additionally includes η_{gate} , the difference between the actual field and F^0 along the reference polynomial. It is bounded by the direct field argument in Proposition 1.2, using neuronwise input and output excursions

$$d_i^u = a\|y'_{u_i}(0)\| + \frac{1}{2}a^2\|y''_{u_i}(0)\| + \frac{1}{6}a^3\|y'''_{u_i}\|,$$

and the analogous d_i^w . The corresponding whole-parameter excursion bounds residual changes. Thus $\eta = \eta_{\text{sm}} + \eta_{\text{gate}}$ is uniform on the step, including all reference-path gate changes.

For the one-sided rate on a tube, let A_i^0 be its midpoint residual matrices and D_* the pointwise output-change coefficient from the midpoint throughout the tube. If p_{in} flags every potentially changed gate there, a valid smooth-piece bound is

$$a_* = \max_i \|A_i^0\| + \lambda D_* + \max_i N^{-1} \sum_n p_{in} \|e_y(x_n)\| \|x_n\|.$$

For gate jumps the validator bounds the positive conormal residual throughout the same tube, sorts lower bounds on its reference-path gate distances, and sums upward-rounded weights. With $\beta = 1$, the intercept $\max_s (B_i(s) - s)_+$ only requires those finite sorted distances. All tube radii are explicit inputs, rather than inferred from successful numerical integration. Corollary 1.4 is checked at every step. Its strict inequality is what validates the reference, independently of the integration method originally used to choose its coefficients.

2 A sharper one-sided gate bound

Let $y = (u_i, w_i)_i$, F be the empirical ReLU gradient field, and $e_y(x) = x - f_y(x)$. For a radius R , define the potentially crossing samples $\mathcal{N}_i = \{n : |u_i^T x_n| \leq R\|x_n\|\}$ and

$$C_i = \frac{1}{N} \sum_{n \in \mathcal{N}_i} \|e_y(x_n)\| \|x_n\|, \quad K_i = \frac{1}{N} \sum_{n \in \mathcal{N}_i} \|x_n\|^2, \quad D = \|y\| + R/2.$$

Use reference-only positive conormal weights

$$B_i(s) = \frac{1}{N} \sum_{|u_i^T x_n| \leq s \|x_n\|} (w_i^T e_y(x_n))_+ \|x_n\|.$$

Suppose $B_i(s) \leq a_i + \beta s$ for $0 \leq s \leq R$, and the residual Hessian matrices on smooth pieces of this ball have norm at most A . Then, for compatible selectors and $\|z\| \leq R$,

$$\langle z, F(y+z) - F(y) \rangle \leq (A + \beta + \Gamma) \|z\|^2 + \|(a_i)_i\|_2 \|z\|, \quad (3)$$

where

$$\Gamma = \frac{1}{2} \max_i (C_i + DRK_i) + D \left(\sum_i \|w_i\|^2 K_i^2 \right)^{1/2}.$$

In particular parameter perturbations need not be charged independently by R inside every neuron's positive gate weight.

Proof. On each smooth piece $DF = -J^*J + H_e$; retain the nonpositive term and bound the residual-Hessian term by $A\|z\|^2$ along the parameter segment. At a gate crossing on this segment the contribution to the scalar pairing is at most

$$\frac{|z_{u_i}^T x_n|}{N} (w_i(s)^T e_{y+sz}(x_n))_+.$$

The reference term gives $\sum_i \|z_{u_i}\| B_i(\|z_{u_i}\|)$. Write $r = \|z\|$. ReLU Lipschitzness along the parameter segment gives $\|f_{y+sz}(x_n) - f_y(x_n)\| \leq Dr\|x_n\|$. The other conormal terms are bounded by

$$\|z_{w_i}\| \|e_y(x_n)\| + \|w_i\| Dr\|x_n\| + \|z_{w_i}\| Dr\|x_n\|.$$

After summing over possible crossings their contribution is at most

$$\sum_i (C_i + DrK_i) \|z_{u_i}\| \|z_{w_i}\| + Dr \sum_i \|w_i\| K_i \|z_{u_i}\| \leq \Gamma r^2.$$

Use $2ab \leq a^2 + b^2$, Cauchy-Schwarz, and $r \leq R$. The reference weights contribute at most $\|(a_i)\|r + \beta r^2$. Endpoint partial selectors and simultaneous crossings satisfy the same bound. $\square$

Signed enclosure of the residual matrices. A second improvement avoids adding a full absolute near-gate contribution to a matrix that already contains it. At a reference midpoint put $c_{n,ab} = e_y(x_n)_a x_{n,b}$. For an uncertain gate set $\mathcal{N}_i$ and the certainly active set $\mathcal{F}_i$, the matrix computed with reference residual and actual selectors belongs entrywise to

$$\left[\frac{1}{N} \sum_{n \in \mathcal{F}_i} c_{n,ab} + \frac{1}{N} \sum_{n \in \mathcal{N}_i} \min(0, c_{n,ab}), \quad \frac{1}{N} \sum_{n \in \mathcal{F}_i} c_{n,ab} + \frac{1}{N} \sum_{n \in \mathcal{N}_i} \max(0, c_{n,ab}) \right].$$

Residual change costs at most λD_{out} in operator norm if $\|f_\theta - f_y\|_n \leq \sqrt{\lambda} D_{\text{out}}$ and $\|\mathbb{E}_n x x^T\| \leq \lambda$. This follows by Cauchy-Schwarz, retaining the signed coordinate intervals before taking a matrix norm. Both estimates are deterministic inequalities. The initialized interval validator implements them with outward bounds; its floating-point feasibility prototype is separate and is not used as a proof premise.

3 The first initialized training-gate event

This result concerns the exact binary input dataset and exact archived initial parameter rows. No numerical trajectory is a proof premise. Indices in the following statement are zero based.

Proposition 3.1 (An isolated transverse first event). *The first training preactivation to vanish is that of neuron 36 on sample 3971. Its time τ satisfies*

$$35888/10^{10} < \tau < 35891/10^{10}.$$

The gate changes from inactive to active. Both one-sided normal velocities are greater than 0.0006064. All other training gates stay strictly in their initial signs through the corresponding frozen-mask comparison interval. The initialized Clarke flow is unique up to this event and has a unique local continuation across this isolated transverse crossing.

Proof. The interval computation resolves every initial gate strictly. Extend the initial-mask vector field polynomially and call its unique smooth solution $y(t)$. Work in the radius- 10^{-6} ball about the archived initial state. The certified speed is at most 0.108613, so on $[0, H]$, $H = 35891/10^{10}$, a first-exit argument confines y to the ball. The field Lipschitz and third-trajectory-derivative bounds are constructed from the full empirical covariance and initial residual. In particular $\|y'''\| < 3.20818$ throughout this interval.

The code evaluates the degree-two Taylor polynomial

$$p(t) = y(0) + tF(y(0)) + \frac{1}{2}t^2DF(y(0))F(y(0))$$

with outward interval arithmetic and adds $3.20818t^3/6$ as a norm remainder. For the candidate preactivation, its upper bound at the lower time is $-5.3279 \cdot 10^{-14}$, and its lower bound at the upper time is $1.2953 \cdot 10^{-13}$. Its derivative is bounded below by 0.000606439. Every other oriented preactivation is positive throughout $[0, H]$: its initial value plus the negative part of its initial slope times H , minus the uniform quadratic remainder, is greater than $2.6079 \cdot 10^{-9}$. Thus the frozen flow has exactly this first event, and coincides with the initialized flow until it occurs.

At the event the only field jump is the input-coordinate vector $N^{-1}(w_i^T e_x)x_n$. Outward bounds on the same ball show that adding its normal projection leaves the post-event normal velocity greater than 0.000606438. Both fields therefore cross the isolated hyperplane in the same direction. Smooth uniqueness on either side and strict transversality exclude dwelling or a choice of distinct continuations at the boundary. The full numerical inequalities, exact rational time endpoints, and input hash are in `FIRST_EVENT_CERTIFICATE.json`. $\square$

This validates one nonlinear event, not a sequence of all subsequent events. It does not establish initialized entry at time 3.4.

4 Initialized continuation and the failed entry comparison

The initialized reversal theorem remains open. Starting at the exact archived parameter vector, the interval verifier establishes a tube through 4974 intervals of length $1/5000$, hence through $t = 0.9948$. The initial comparison radius is exactly zero. Every step satisfies its strict tube inequality, including all compatible training-gate changes. All smooth-rate coefficients were independently recomputed, and the accumulation was replayed from zero with 60-digit interval exponentials.

Let $\bar{r}_j$ be the outward-rounded comparison radius, $k_j \geq 0$ its one-sided rate, and $\omega_j \geq 0$ the sum of the reference-defect and adverse-gate intercept bounds. The checked recurrence is

$$\bar{r}_j = \text{up} \left[e^{k_j/5000} \bar{r}_{j-1} + \frac{e^{k_j/5000} - 1}{k_j} \omega_j \right].$$

At the first failure of the *entry-budget* comparison, the ledger gives

$$\begin{aligned} j &= 4974, & t_{j-1} &= 0.9946, & t_j &= 0.9948, \\ \bar{r}_{j-1} &\approx 1.1982855683396031 \cdot 10^{-5}, \\ \bar{r}_j &\approx 1.2003084494454761 \cdot 10^{-5} > 1.2 \cdot 10^{-5}. \end{aligned}$$

Here $k_j < 4.709629$, $\omega_j < 4.466164 \cdot 10^{-5}$. The prescribed tube radius is greater than $1.5425235 \cdot 10^{-5}$; its strict slack is greater than $3.4221508 \cdot 10^{-6}$. Thus the actual tube criterion has *not* failed. The failed sufficient inequality is specifically

$$\boxed{\bar{r}_{4974} \leq 1.2 \cdot 10^{-5}}. \tag{4}$$

The exact binary upper bounds and rational step are recorded in `INITIAL_FLOW_CERTIFICATE.json`; the full accepted prefix is in `INITIAL_FLOW_ENCLOSURE.csv`.

This distinction is essential. The result proves $\|\theta(t_j) - y(t_j)\| \leq \bar{r}_j$, not a lower bound on the actual error. The nondecreasing comparison used here cannot subsequently restore the selected entry tolerance at $t = 3.4$. A sharper propagation method, a better-defect reference, a different sufficient entry set, or an observable enclosure could still succeed. No structural obstruction to initialized reversal, and no justification for a theorem-scope pivot, follows from (4).

At this step the reference gate correction is less than $4.068304 \cdot 10^{-5}$, while the smooth reference defect is less than $1.611855 \cdot 10^{-6}$. The proof bottleneck is therefore not addressed by checking only a smooth integrator's truncation error. A continuation that improves this estimate must account for actual gate events and their signed propagation. The isolated first-event proposition above does this for one event; it does not certify the entire event sequence.

Independent arithmetic audit. A suspected normalization error proved to be a misreading of parentheses. Both the certainly-active and uncertain residual contributions had exactly one factor $1/N$. Independent recomputation agreed within $1.8 \cdot 10^{-15}$ in the smooth rates; the delivered expression makes that normalization explicit. No alleged correction to the underlying mathematical bound is claimed.

5 Timing relation: the remaining signed inequality

No initialized timing theorem is established. The certified endpoint signs alone imply neither a unique zero of Q_G nor an ordering of that zero and the full-network minimum. In particular the interpolated empirical times are not used as theorem constants.

For the actual probe, put $g_E = \nabla_\theta E$ on each probe activation cell, and let $g_G = \nabla \Phi_I$ be the repository statistic's gradient defined above. Almost everywhere along a compatible flow, the exact discrepancy is

$$\Xi(\theta) = (g_E - g_G)^T F, \quad \dot{E} = Q_G + \Xi.$$

Thus one concrete sufficient signed estimate after a reached G crossing is

$$0 \leq Q_G \leq q_+, \quad \Xi \leq -q_+ - \delta, \quad \delta > 0. \quad (5)$$

This would force the full error still to decrease on that interval. All neurons, the ungated fixed cohort defining Q_G , and actual training selectors must be retained in this comparison. At probe-gate events one must use the actual one-sided/a.e. derivative rather than extending a single probe-cell gradient across the event.

To localize all full-error minimizers to $[a_E, b_E]$, it suffices additionally to certify $\dot{E} < 0$ almost everywhere from entry to a_E , and $\dot{E} > 0$ almost everywhere from b_E to the final endpoint. A reached sign-separated G crossing bracket $[a_G, b_G]$ with $b_G < a_E$ would then prove a positive timing separation without asserting uniqueness of either crossing. These interval-wide inequalities have not been verified. Per the requested dependency order, no timing search was substituted for the missing initialization-to-entry proof.